THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 11: **Earth, Moon and the Sun** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A farmer relies on the lunar calendar for the harvest and a friend on the solar calendar. The solar calendar tracks: 1

(a) eclipses only **(b)** phases only **(c)** seasons well **(d)** tides only

2. If the Moon did not rotate, we would see: 1

(a) only dark side **(b)** only one face

(c) all of its surface over a month **(d)** no face

3. A place on the equator has a shadow at noon that is: 1

(a) longest on equinoxes **(b)** constant

(c) absent always **(d)** shortest on equinoxes

4. Which of the following would not change if Earth's axis tilt were zero? 1

(a) length of day variation **(b)** time taken for one rotation

(c) seasons **(d)** solstices

5. Earth takes 365.25 days to orbit. In 100 years the number of leap days (approx.) is: 1

(a) 100 **(b)** 10 **(c)** 25 **(d)** 50

6. Which of these Moon phases occurs when the Earth is between the Sun and the Moon? 1

(a) full moon **(b)** first quarter **(c)** new moon **(d)** last quarter

7. A sundial shows noon when the gnomon's shadow points: 1

(a) west **(b)** north (in the northern hemisphere)

(c) south **(d)** east

8. At a place on the Arctic Circle, the Sun does not set for about: 1

(a) a day at summer solstice **(b)** never

(c) six months **(d)** a week

9. **Assertion (A):** A solar eclipse occurs on a new moon day. 1

Reason (R): During a solar eclipse, the Earth comes between the Sun and the Moon.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** A leap year has 366 days. 1
Reason (R): The Earth takes about $365\frac{1}{4}$ days to go around the Sun, so the extra quarter days add up to one day in four years.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. What is the benefit of the lunar calendar in festivals? 2

12. How can you prove that the Moon rotates though we see one face? 2

13. Why are the Moon's phases the same for all people on Earth on a given night? 2

14. Why is the sunrise time different in different cities? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how seasons would differ in the southern hemisphere in the month of June. 3

16. Explain why the Moon is visible in the day sometimes but not Venus. 3

17. Explain the relationship between the Moon and the Indian calendar (Purnima, Amavasya). 3

OR

Calculate the number of days between full moon on 12 Jan and the next one, taking the lunar month as 29.5 days. 3

18. Explain why 21 June is not the hottest day in India. 3

OR

Explain why the Moon looks about as big as the Sun although the Sun is much larger. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** During a solar eclipse, Rani's school arranges safe viewing with special filters. A week later, a lunar eclipse is seen and Rani notices that the Moon appears dark red. 5
(i) Which body lies between the other two during a solar eclipse?
(ii) On which Moon-day does a solar eclipse occur?
(iii) Why should we never look at the Sun directly during an eclipse?
(iv) During a lunar eclipse, whose shadow falls on which body, and on which Moon-day does it occur?

20. Analyse the statement 'If the Earth had no Moon, nights would be darker and eclipses would not occur.' 5

OR

Discuss the cultural importance of the Sun, Moon and eclipses in India and how scientific understanding helped remove myths. 5

END OF QUESTION PAPER — Total: 40 Marks